

A NEGATIVE ANSWER TO MOHAMMADI'S QUESTION ON POWERS OF COVER IDEALS

ABSTRACT. We exhibit an eight-vertex Cohen–Macaulay graph G whose cover ideal $J(G)$ has a 5-linear resolution over every field, while its ordinary square does not have a linear resolution. In fact,

$$\beta_{1,(1,1,2,2,2,2,1,1)}(J(G)^2) = 1.$$

This gives a negative answer to a question of Mohammadi on powers of cover ideals.

Let K be a field and $S = K[x_1, \dots, x_8]$. For a graph H on $[8]$, write

$$I(H) = (x_i x_j : \{i, j\} \in E(H)), \quad J(H) = I(H)^\vee = \bigcap_{\{i,j\} \in E(H)} (x_i, x_j).$$

Thus $J(H)$ is generated by the monomials corresponding to the minimal vertex covers of H . A graph is Cohen–Macaulay over K when $S/I(H)$ is Cohen–Macaulay. We write

$$\beta_{i,\mathbf{b}}(M) = \dim_K \operatorname{Tor}_i^S(M, K)_{\mathbf{b}}.$$

Mohammadi asked whether every ordinary power of the cover ideal of a Cohen–Macaulay graph again has a linear resolution [7, Question 4.1]; by the Eagon–Reiner theorem [1], a graph is Cohen–Macaulay over K precisely when its cover ideal has a linear resolution over K . Hà and Van Tuyl restate the question as [3, Question 6.2]:

Let G be a graph such that the cover ideal $J(G)$ has a linear resolution. Does $J(G)^s$ have a linear resolution for all $s \geq 1$? If not, what hypotheses are needed on G to give this conclusion?

Mohammadi's positive result for cactus graphs was later supplied with a corrected proof by Lu and Wang [5]. The following example answers the general question in the negative already for the square.

Theorem 1. *Let G be the graph on $[8]$ with*

$$E(G) = \{14, 16, 17, 23, 26, 28, 37, 48, 56, 57, 58\},$$

where ij denotes the edge $\{i, j\}$. Then G is Cohen–Macaulay over every field, $J(G)$ has a 5-linear resolution, and

$$\beta_{1,\mathbf{a}}(J(G)^2) = 1, \quad \mathbf{a} = (1, 1, 2, 2, 2, 2, 1, 1).$$

Consequently, $J(G)^2$ does not have a linear resolution over any field.

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Proof. For distinct indices, write $i_1 \cdots i_r$ for $\{i_1, \dots, i_r\}$. Splitting the maximal independent sets of G according to their intersection with $\{3, 4\}$ gives

$$\begin{array}{ll} \text{both } 3, 4: & 345, 346; \\ 3 \text{ but not } 4: & 135, 138, 368; \\ 4 \text{ but not } 3: & 245, 247, 467; \\ \text{neither:} & 125, 678. \end{array}$$

Hence $\Delta = \text{Ind}(G)$ has exactly $2 + 3 + 3 + 2 = 10$ facets, each of size 3. Reordered, they are

$$(1) \quad 345, 135, 245, 125, 346, 247, 467, 138, 368, 678.$$

For the second through tenth facets, the facets of the intersection with the preceding subcomplex are, respectively,

$$35; \quad 45; \quad 15, 25; \quad 34; \quad 24; \quad 46, 47; \quad 13; \quad 36, 38; \quad 67, 68.$$

Each intersection is nonempty and pure of dimension one. Hence (1) is a shelling of the pure two-dimensional complex Δ . Thus

$$K[\Delta] = S/I(G)$$

is Cohen–Macaulay over every field. The Eagon–Reiner theorem [1] now implies that $J(G) = I(G)^\vee$ has a linear resolution. Since every facet of Δ has size 3, every minimal vertex cover has size 5, so this resolution is 5-linear.

Let C_i be the complement in $[8]$ of the i th facet in (1). The minimal vertex covers are

$$\begin{array}{llllll} C_1 = 12678, & C_2 = 24678, & C_3 = 13678, & C_4 = 34678, & C_5 = 12578, \\ C_6 = 13568, & C_7 = 12358, & C_8 = 24567, & C_9 = 12457, & C_{10} = 12345. \end{array}$$

Set $x_C = \prod_{i \in C} x_i$ and

$$m = x^{\mathbf{a}} = x_1 x_2 x_3^2 x_4^2 x_5^2 x_6^2 x_7 x_8.$$

Consider the upper Koszul complex

$$\mathcal{K}^{\mathbf{a}} = \left\{ F \subseteq [8] : \frac{m}{x_F} \in J(G)^2 \right\}.$$

Since $\mathbf{a}$ has all coordinates positive, m/x_F is a monomial for every $F \subseteq [8]$, and since $J(G)^2$ is an ideal, $m/x_{F'} = (m/x_F) x_{F \setminus F'}$ lies in $J(G)^2$ whenever $F' \subseteq F$ and $F \in \mathcal{K}^{\mathbf{a}}$. Thus $\mathcal{K}^{\mathbf{a}}$ is closed under passing to subsets, hence is a simplicial complex.

The identities

$$(2) \quad \frac{m}{x_3 x_4} = x_{C_6} x_{C_8}, \quad \frac{m}{x_5 x_6} = x_{C_4} x_{C_{10}}$$

show that 34 and 56 are faces.

Since $J(G)^2$ is a monomial ideal, membership is equivalent to divisibility by $x_{C_i} x_{C_j}$ for some $1 \leq i \leq j \leq 10$. The following table excludes four

vertices and the four cross-edges between $\{3, 4\}$ and $\{5, 6\}$:

(3)

F	condition on each factor cover	eligible covers	t
$\{1\}$ or $\{3, 5\}$	omit 1 or contain 4, 6	C_2, C_4, C_8	7
$\{2\}$ or $\{4, 5\}$	omit 2 or contain 3, 6	C_3, C_4, C_6	8
$\{7\}$ or $\{4, 6\}$	omit 7 or contain 3, 5	C_6, C_7, C_{10}	1
$\{8\}$ or $\{3, 6\}$	omit 8 or contain 4, 5	C_8, C_9, C_{10}	2

For a singleton $F = \{r\}$ occurring in (3), the x_r -exponent of m/x_F is zero, so both factor covers must omit r . For a two-element F in the table, m/x_F and $x_{C_i}x_{C_j}$ both have degree 10; divisibility would therefore be equality, and every exponent-two coordinate of m/x_F must occur in both covers. In each row, the two conditions correspond respectively to the two choices of F . These observations give the eligible covers in (3). In each row every eligible cover contains t , so a product of two such covers has x_t -exponent 2, whereas m/x_F has x_t -exponent 1. Thus none of the eight displayed faces belongs to $\mathcal{K}^{\mathbf{a}}$.

Because 1, 2, 7, 8 are not vertices of $\mathcal{K}^{\mathbf{a}}$, every face is contained in $\{3, 4, 5, 6\}$. Any subset of this set not contained in 34 or 56 contains one of the excluded cross-edges 35, 36, 45, 46. Together with (2), this proves

$$\mathcal{K}^{\mathbf{a}} = \langle 34, 56 \rangle,$$

the disjoint union of two edges. The upper-Koszul formula [6, Theorem 1.34] gives

$$\beta_{1,\mathbf{a}}(J(G)^2) = \dim_K \tilde{H}_0(\mathcal{K}^{\mathbf{a}}; K) = 1.$$

Finally, $J(G)^2$ is generated in degree 10, while $|\mathbf{a}| = 12$. A 10-linear resolution would have all first syzygies in degree 11, so $J(G)^2$ is not linear. $\square$

Remark 2. The vertices 1, 6, 5, 7 induce a 4-cycle, so G is not chordal. Thus Theorem 1 does not address the Herzog–Hibi–Ohsugi conjecture that every power of the cover ideal of a chordal graph is componentwise linear [4, Conjecture 2.5], surveyed in [3].

Ficarra and Moradi constructed analogous failures for squarefree monomial ideals in every degree $3 \leq d \leq n-3$ [2, Theorem D], but their construction is not a graph cover ideal. For $d \geq 4$ its generators have a nontrivial common factor, whereas a graph cover ideal has greatest common divisor 1 because every vertex extends to a maximal independent set. For $d = 3$, in their notation,

$$\{a, c, f, y_1, \dots, y_{n-6}\}$$

is a minimal hitting set of the generator supports, so the Alexander dual has a generator of degree $n-3 \geq 3$ and is not quadratic. Theorem 1 gives the graph-cover-ideal counterexample asked for by Mohammadi.

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